Origins and Evolution of Language
Class 4: Communication and cognition in non-humans, prerequisites for iterated learning

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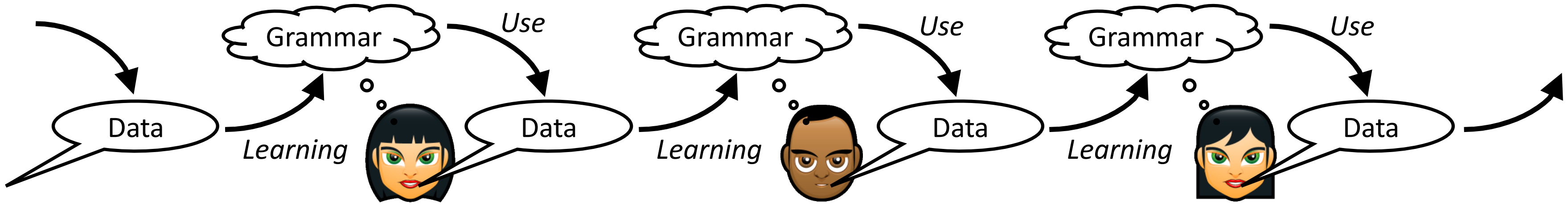
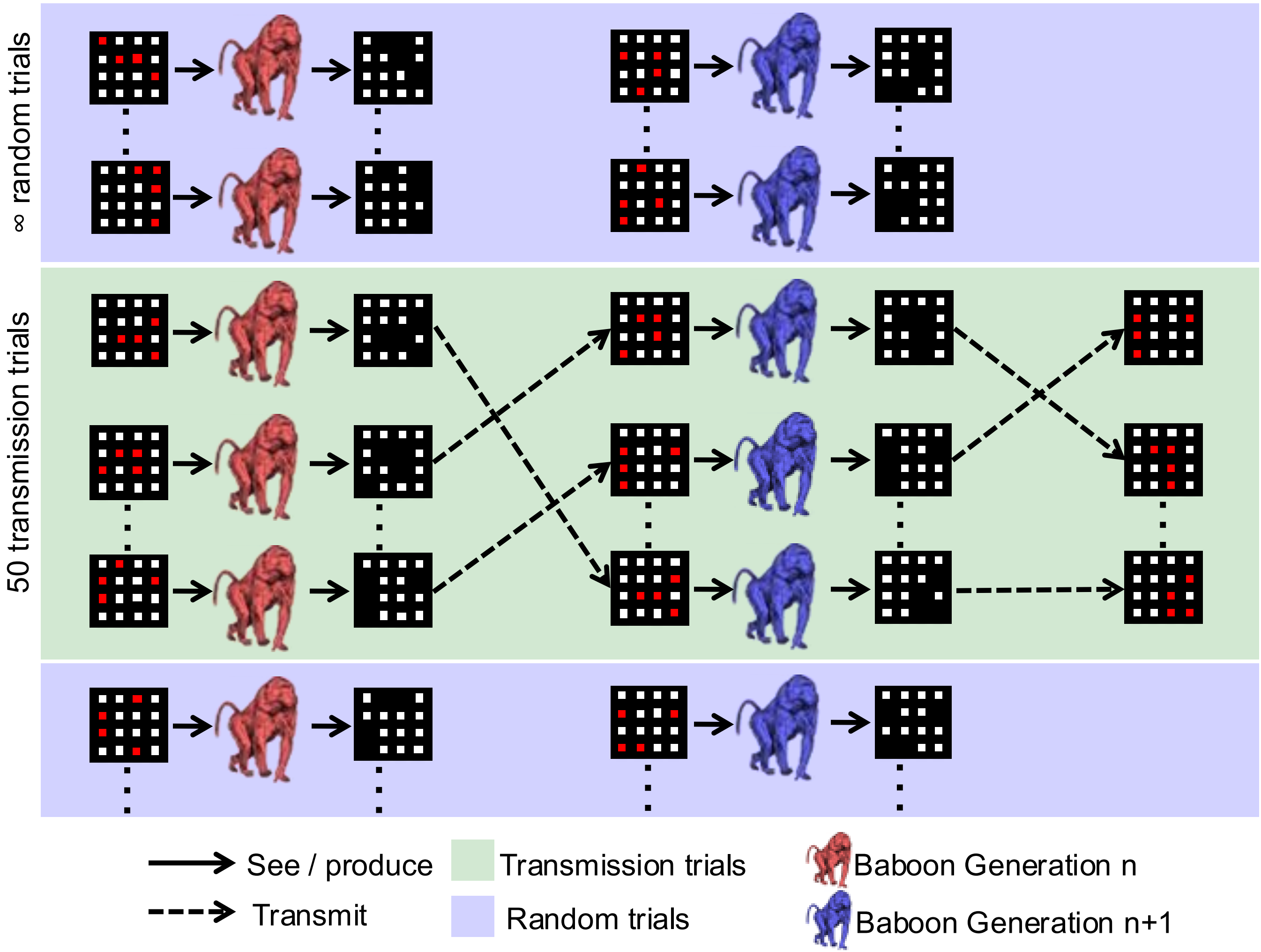
Language is transmitted via repeated **learning** and **use**, and is shaped by these processes

Fundamental "design features" of language (arbitrary symbols, duality of patterning, compositionality) emerge naturally from learning and use

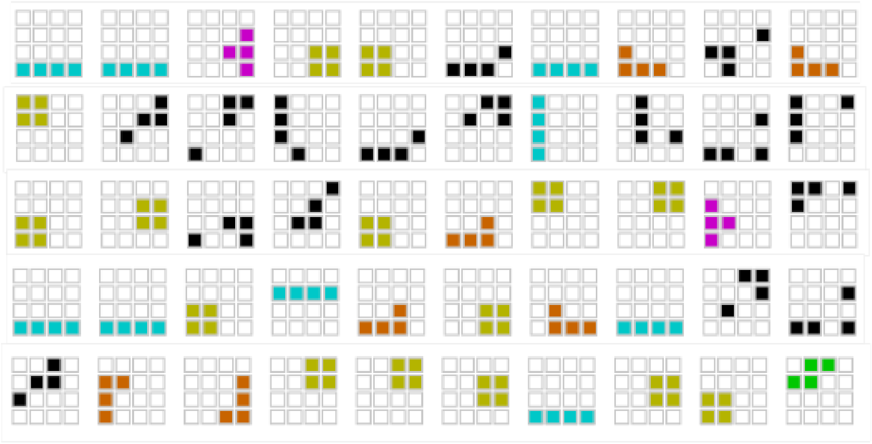
(We should also expect adaptation to different constraints in learning and use to contribute to cross-linguistic variation)



What cognitive prerequisites are required for iterated learning, and how did those evolve?

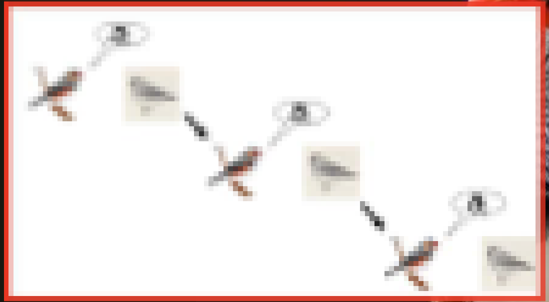

∞ random trials
50 transmission trials
See / produce
Transmit
Transmission trials
Random trials
Baboon Generation n
Baboon Generation n+1

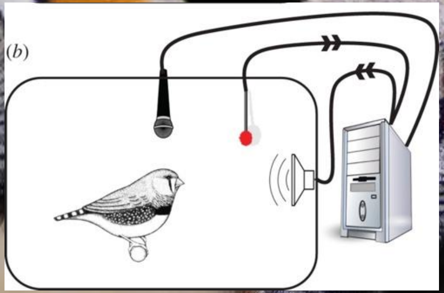
Chain 4, Generation 12

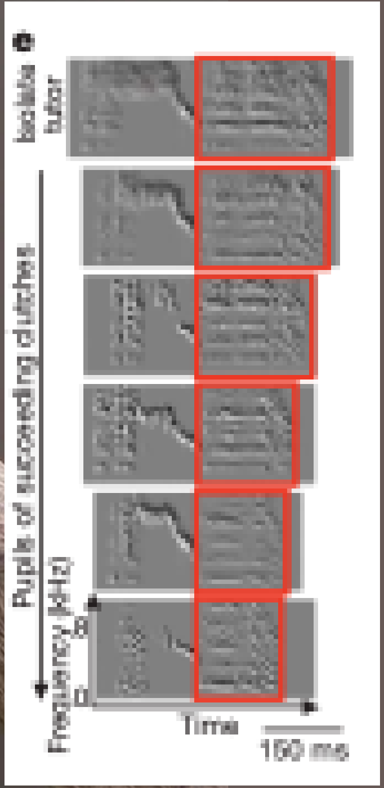


Chain 1, Generation 12

Iterated learning in zebra finches







Zebra finch song evolves from isolate to wild-type rapidly (even under quite minimal self-tutoring)

Fehér, O., et al. De novo establishment of wild-type song culture in the zebra finch. *Nature, 459*, 564–568.
Fehér, O., et al. (2017). Statistical learning in songbirds: from self-tutoring to song culture. *Philosophical Transactions of the Royal Society, 372*, B37220160053.

Dispensing with a couple of old ideas that are probably wrong

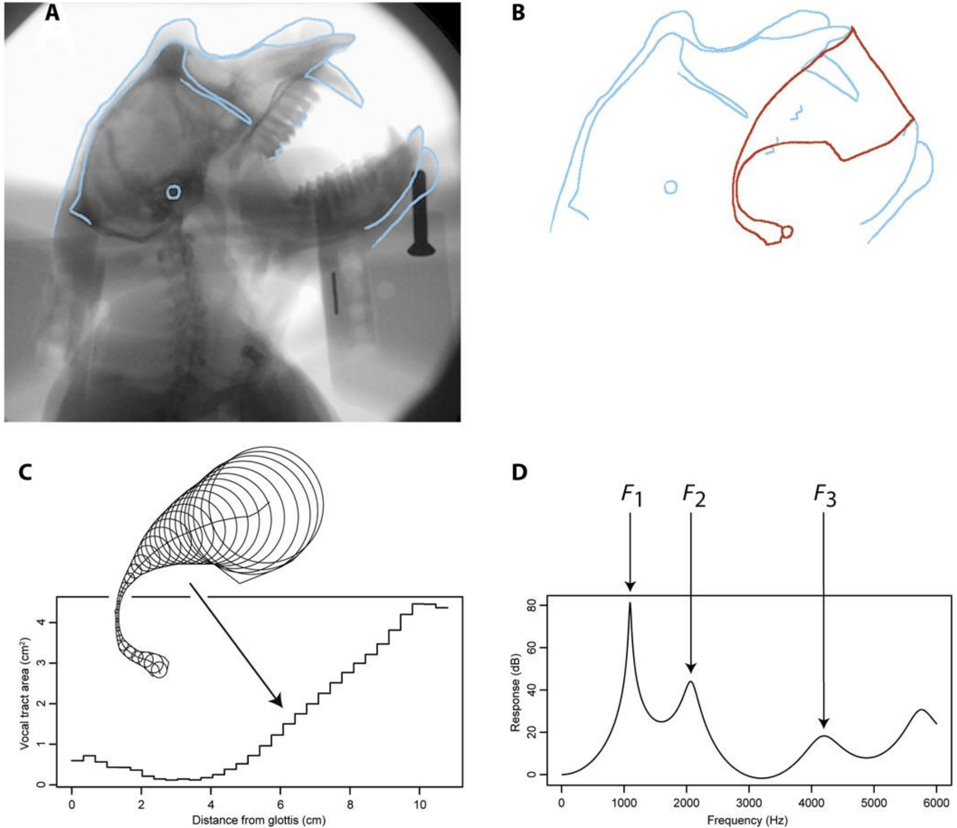
Non-humans have rich mental lives

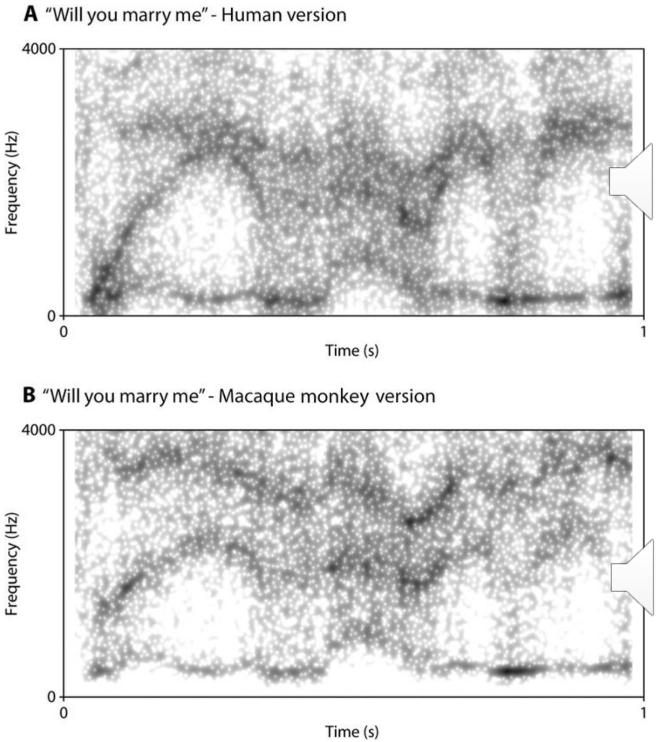
- Concepts and categories
- Memory and planning
- Hierarchically-structured behaviours
- Tool use
- …

(but their communication systems seem quite restricted)

- In the things they communicate about
- 'Innate' signal repertoires
  - Particularly among primates
- "Functionally referential"
  - But not intentional (?)
- Complex vocalisations
  - But less obviously in primates (?)
  - And complexity not subserving meaning (?)

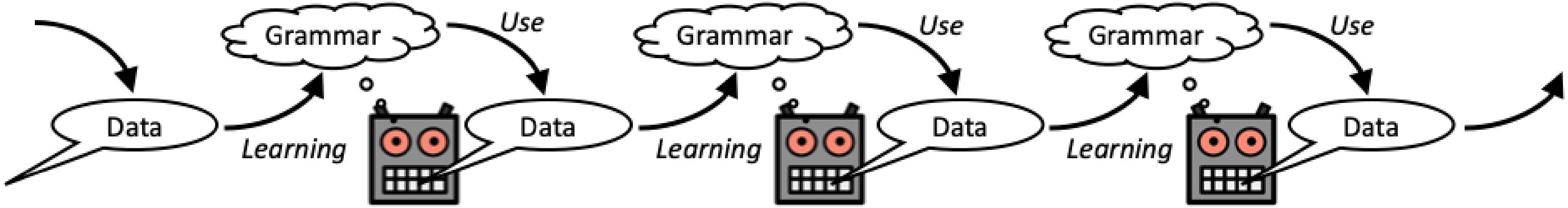
A monkey vocal tract is probably good enough





Fitch, W. T., De Boer, B., Mathur, N., & Ghazanfar, A. A. (2016). Monkey vocal tracts are speech-ready. *Science Advances, 2*, e1600723.

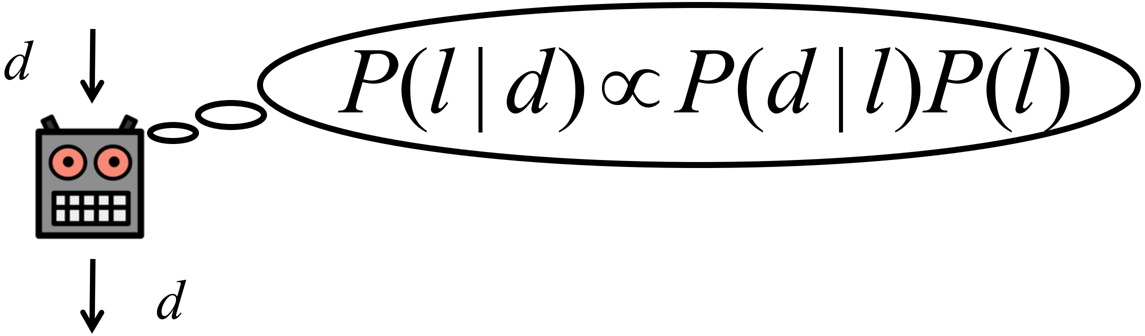
Simulating Iterated Learning



Allows us to check minimal conditions under which these results hold

(Allows us to rule out influence of human participants' native language)

Learning as Bayesian inference



Data: meaning-signal pairs

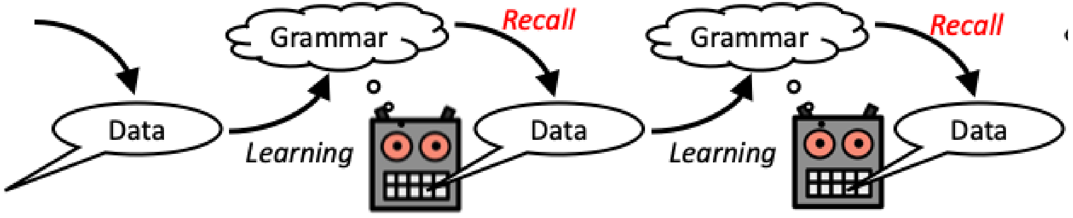
Languages: specify signals for meanings

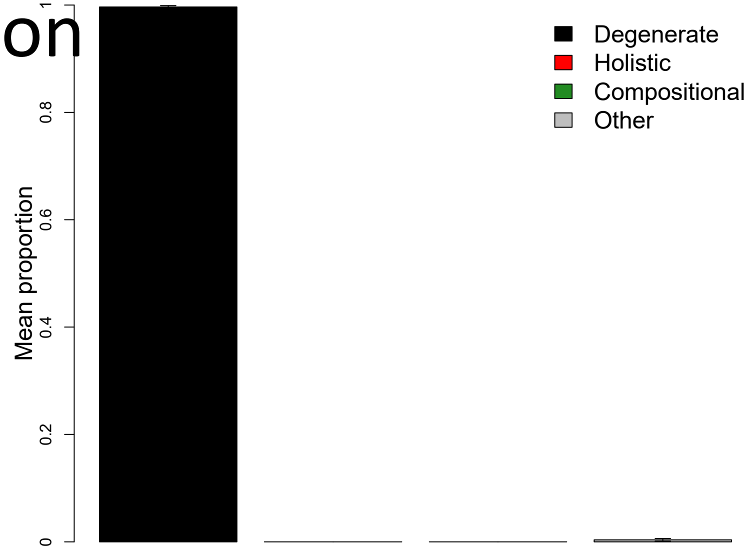
Likelihood p(d|l): avoid ambiguous utterances

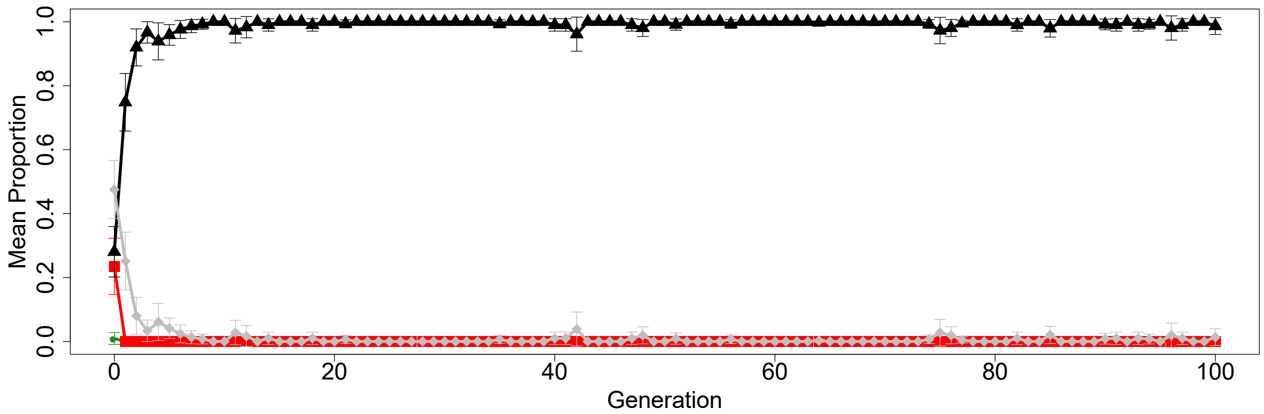
- P(signal) $\propto$ (1/ambiguity)$^{\alpha}$

Prior p(l): learner preference for compressible languages (shorter coding length)

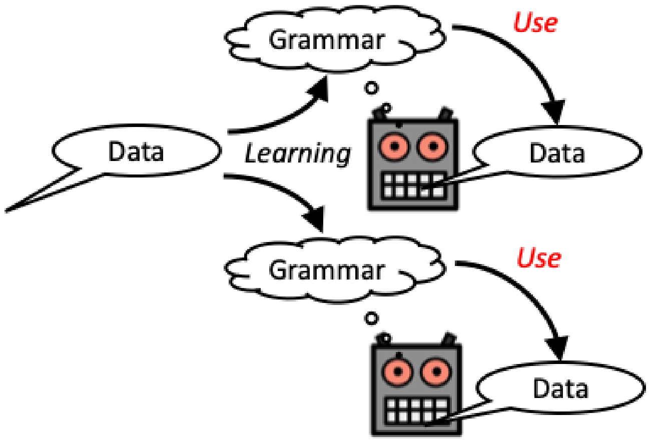
Transmission minus communication

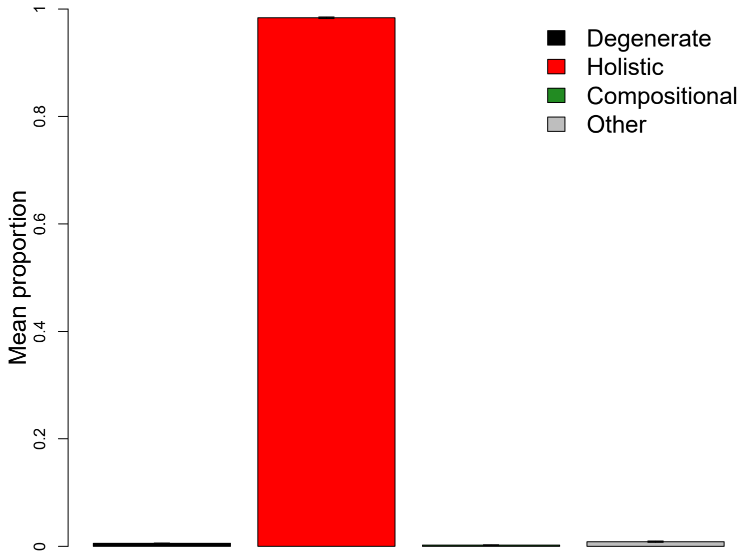


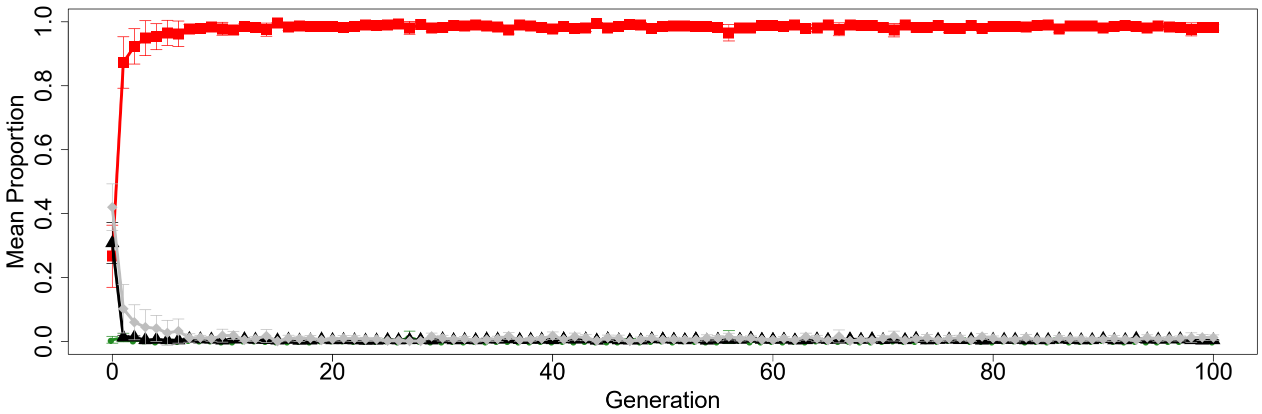




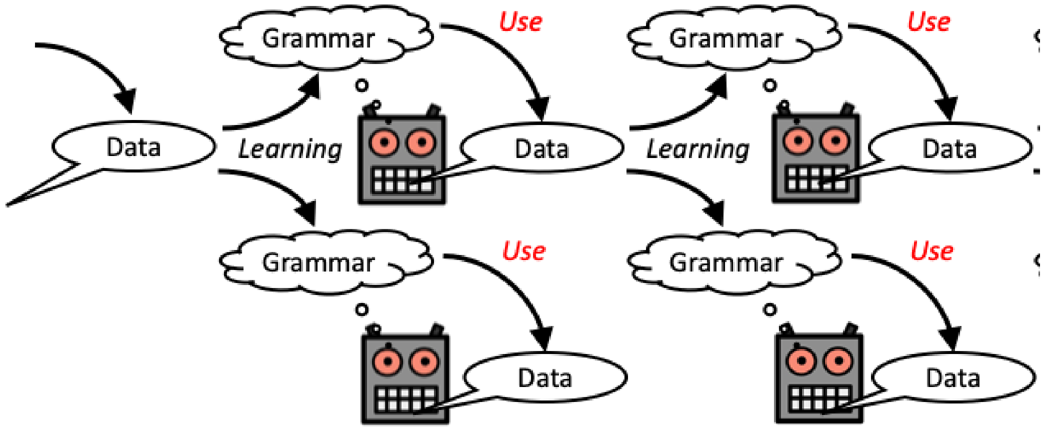
Communication only

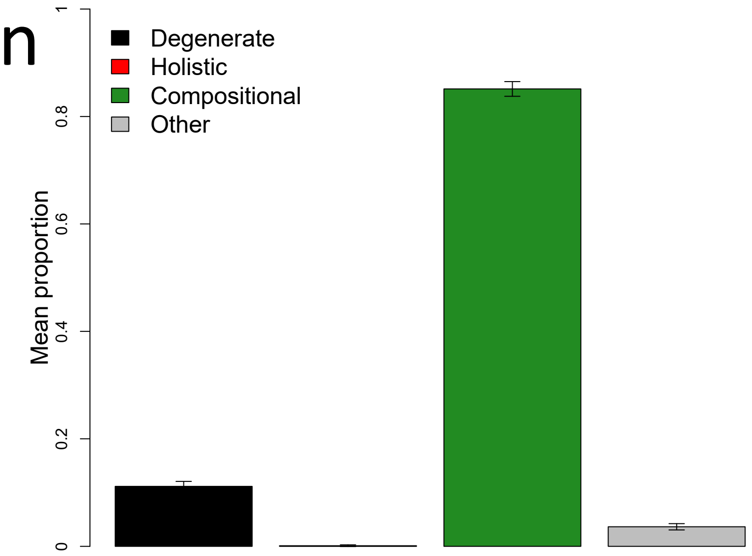


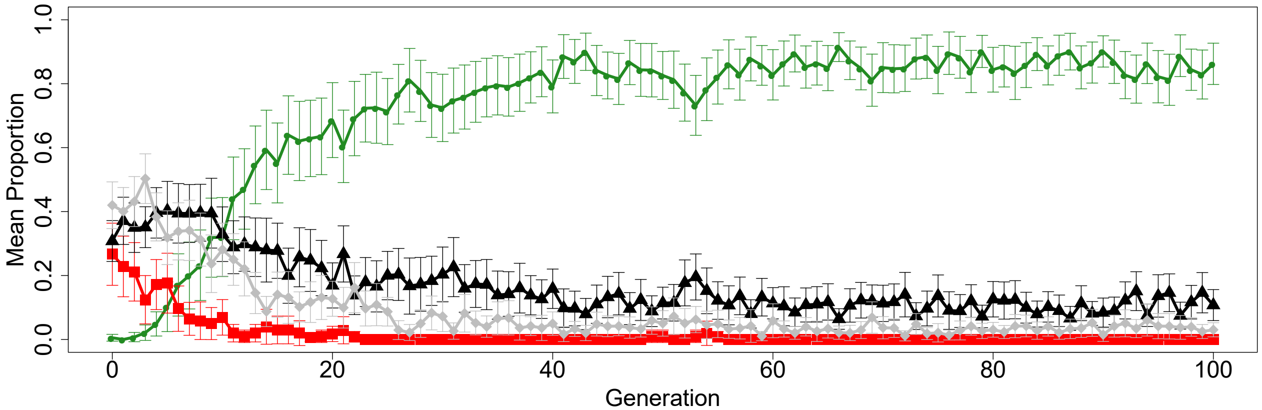




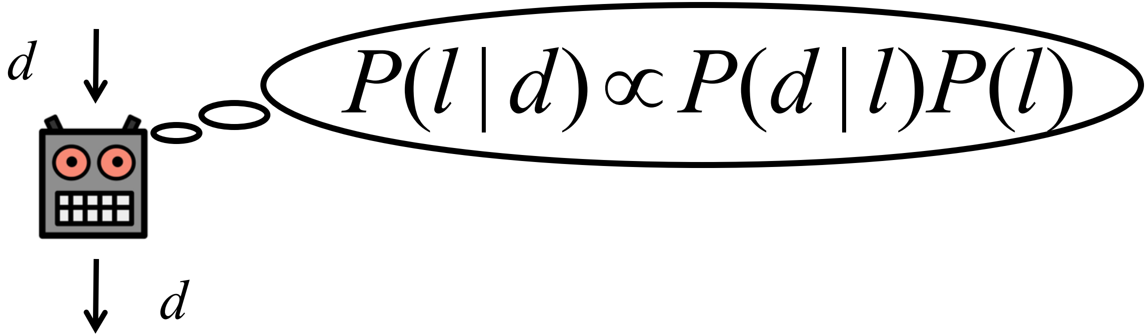
Transmission plus communication







Learning as Bayesian inference



Data: meaning-signal pairs

Access to signal **<u>and its intended meaning</u>**

Languages: specify signals for meanings

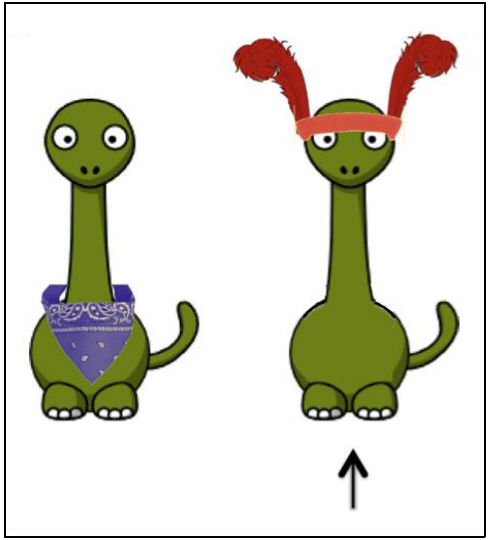
Likelihood p(d|l): avoid ambiguous utterances

- P(signal) ∝ $(1/\text{ambiguity})^{\alpha}$

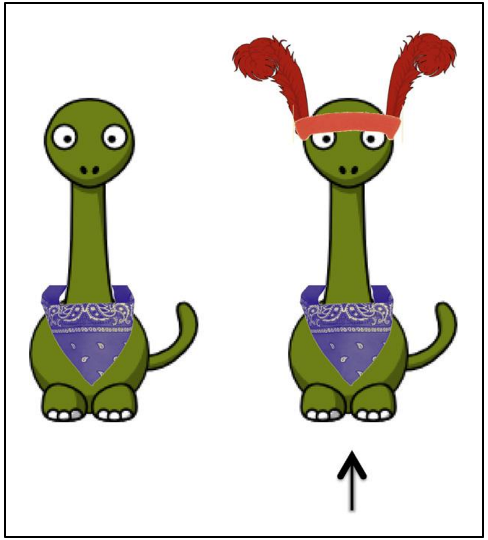
Prior p(l): learner preference for compressible languages (shorter coding length)

Expectations about how people use words

“Filler” (control) trial



Inference trial



“dax” = bandana?

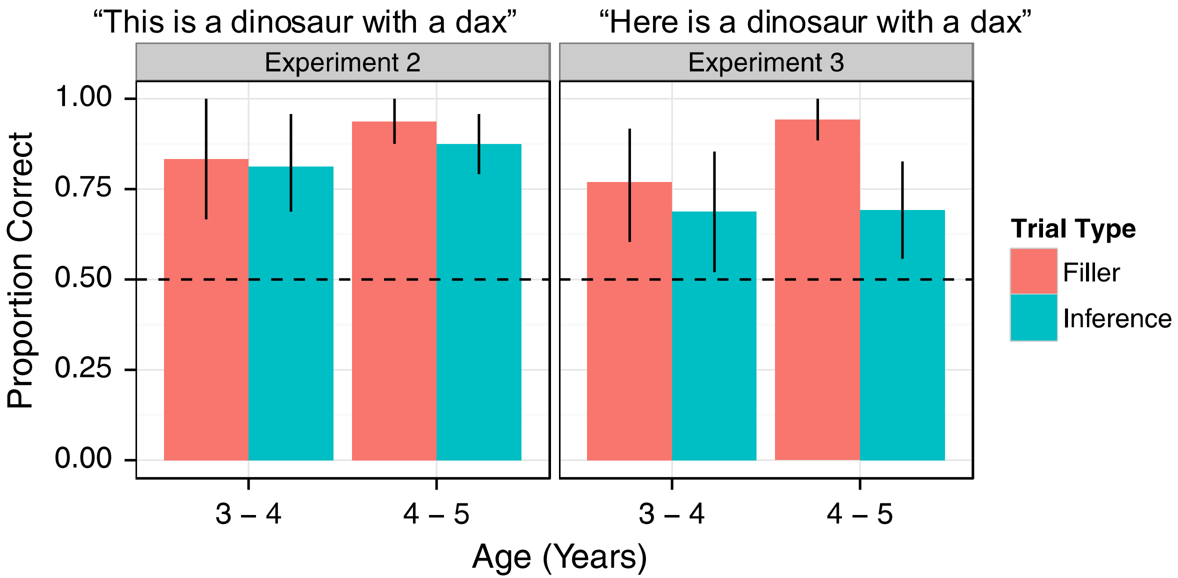
“dax” = head dress?

“This is a dinosaur with a dax” (Exp 2)
“Here is a dinosaur with a dax” (Exp 3)

Frank, M. C., & Goodman, N. D. (2014). Inferring word meanings by assuming that speakers are informative. *Cognitive Psychology, 75,* 80-96.

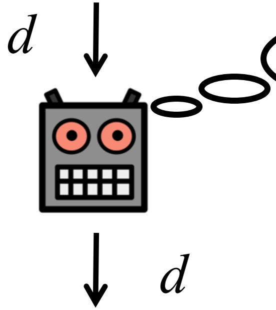
Expectations about how people use words

Children assume that people use words in an informative way



Frank, M. C., & Goodman, N. D. (2014). Inferring word meanings by assuming that speakers are informative. *Cognitive Psychology, 75,* 80-96.

Learning as Bayesian inference



$$P(l \mid d) \propto P(d \mid l) P(l)$$

Data: meaning-signal pairs

Languages: specify signals for meanings

Likelihood p(d|l): avoid ambiguous utterances

- P(signal) $\propto$ (1/ambiguity)$^{\alpha}$

(Simple) pragmatics – model the receiver

Prior p(l): learner preference for compressible languages (shorter coding length)

Gricean / ostensive-inferential communication

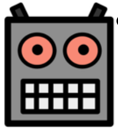
Speaker’s utterances (or other communicative behaviours)

- provide evidence about their thoughts
- are designed to allow the hearer to infer those thoughts

Hearers infer meaning based on these clues and context, guided by the knowledge that the speaker wants the hearer to be able to infer their informative intention

Learning as Bayesian inference

d



d

$$P(l \mid d) \propto P(d \mid l)P(l)$$

Space of reachable languages **includes the structurally interesting ones**!
(sequential constraints, compositional)

Data: meaning-signal pairs

Languages: specify signals for meanings

Likelihood p(d|l): avoid ambiguous utterances

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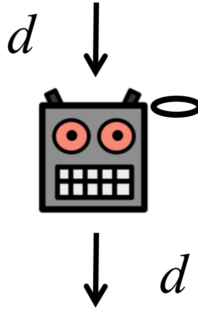
Prior p(l): learner preference for compressible languages (shorter coding length)

Learning as Bayesian inference



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Languages: specify signals for meanings

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- P(signal) ∝ $(1/\text{ambiguity})^{\alpha}$

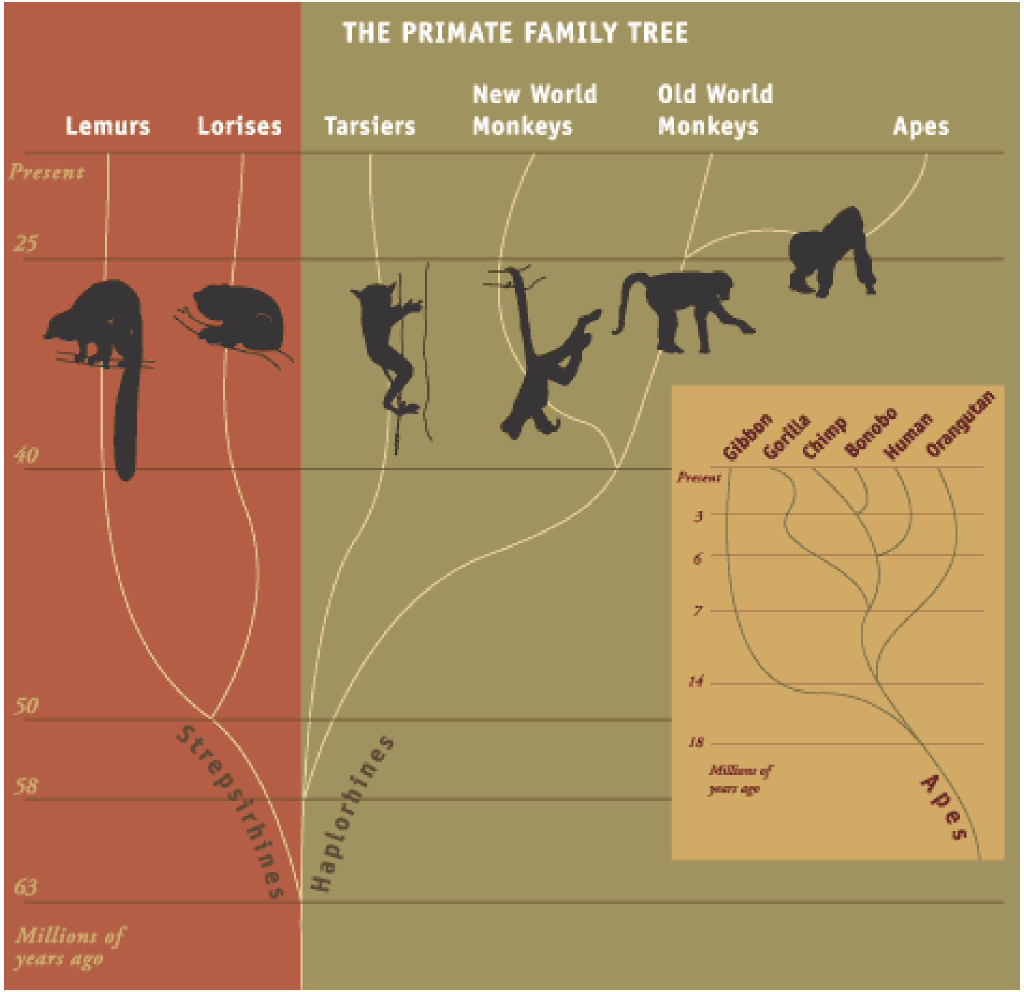
This is probably very general!

Prior p(l): learner preference for compressible languages (shorter coding length)

The human package

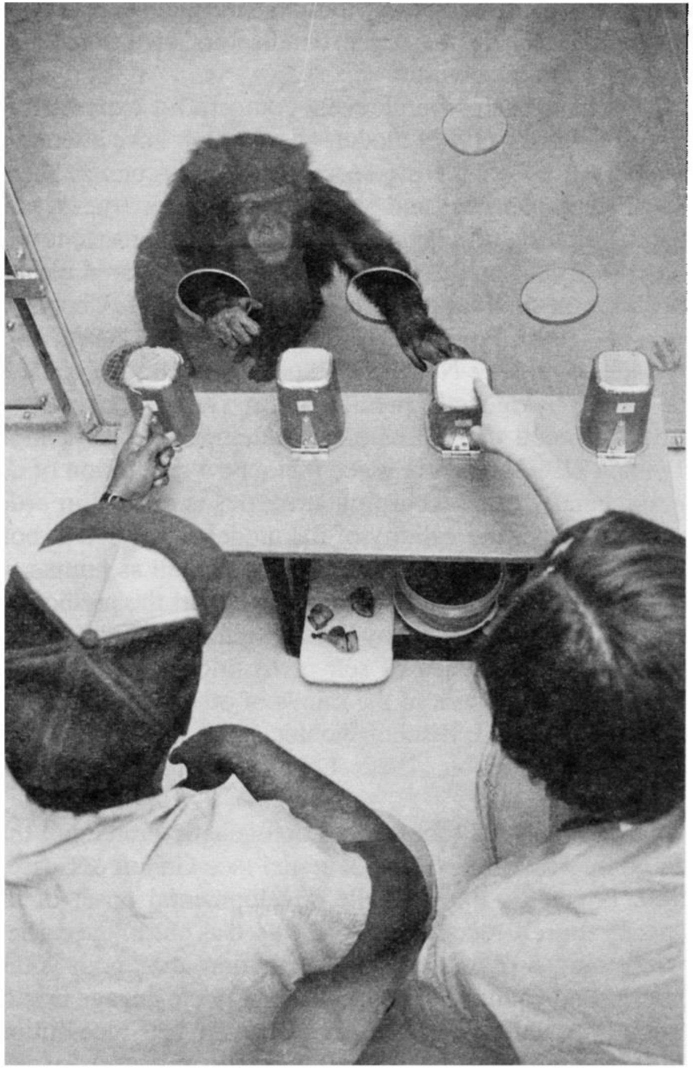
Somehow, we ended up with

- The ability to infer and reason about the minds of others
- The ability to learn complex grammars
  - capacity for complex vocal learning
  - ability to learn complex sequencing constraints
  - ability to learn compositional meaning-form mappings



http://www.daviddarling.info/images/primate_family_tree.gif

Reasoning about other minds



Early view: striking differences

- 6 juvenile chimps (approx. 4 y. o.)
- Two experimenters
- “Guesser” leaves room
- “Knower” hides food under cup
  - Chimp can’t see which one
- Both humans point to a cup
- Chimp indicates which cup he wants to look under

4 y. o. humans can do this, **chimps at chance**

Povinelli, D. J., Rulf, A. B., & Bierschwale, D. T. (1994). Absence of knowledge attribution and self-recognition in young chimpanzees (*Pan troglodytes*). *Journal of Comparative Psychology, 108,* 74–80.

False-belief 1
Chimpanzee Hatsuka

Krupenye, C., et al. (2016). Great apes anticipate that other individuals will act according to false beliefs. *Science, 354,* 110-114.

Familiarization
Bonobo Jasongo

Krupenye, C., et al. (2016). Great apes anticipate that other individuals will act according to false beliefs. *Science, 354,* 110-114.

Knowing others' minds: false belief

Table 1. Number of participants who made first looks to either the target or the distractor during the agent's approach in experiments one (N = 40) and two (N = 30). Values in parentheses indicate the number of participants who did not look at either.

Condition	Target	Distractor	Total
Experiment one			
FB1	10	4	14 (6)
FB2	10	6	16 (4)
Total	20	10	30 (10)
Experiment two			
FB1	8	2	10 (6)
FB2	9	3	12 (2)
Total	17	5	22 (8)

Krupenye, C., et al. (2016). Great apes anticipate that other individuals will act according to false beliefs. *Science, 354,* 110-114.

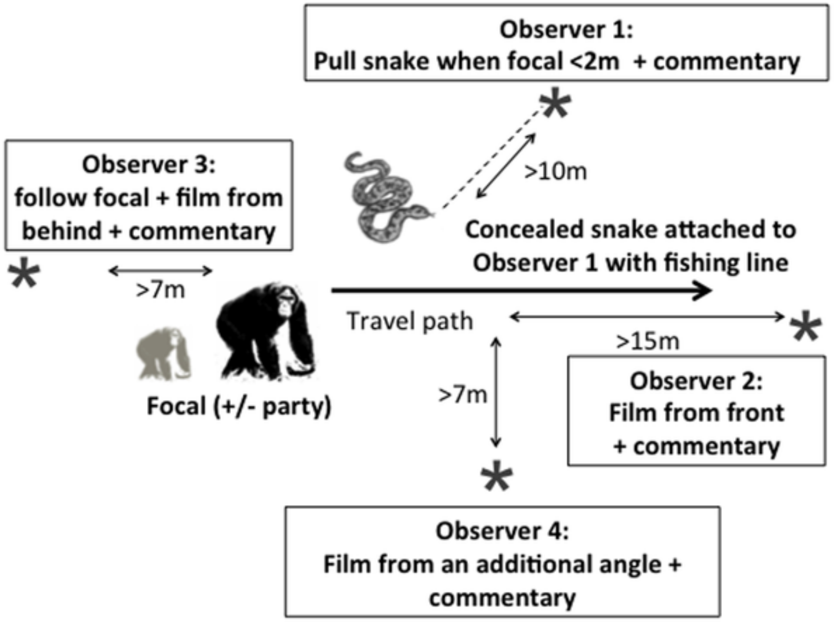
Absence of intentional communication in macaques?

- Mothers and infants
- **Ignorance condition**: Mother knows something, infant doesn't
  - Presence of food, predator
- **Knowledge condition:** They both know it
- **Mothers' vocalizations didn't differ between conditions**

Cheney, D., & Seyfarth, R. (1990). Attending to behaviour versus attending to knowledge: examining monkeys' attribution of mental states. *Animal Behavior, 40,* 742-753.

Intentional communication in chimpanzees?

- Wild chimps
- Surprised with snake model, either alone or in part of group
  - Presence of others matters?
  - Gaze-alternation?
  - Persist until others safe?



Schel, A. M., Townsend, S. W., Machanda, Z., Zuberbühler, K., & Slocombe, K. E. (2013) Chimpanzee Alarm Call Production Meets Key Criteria for Intentionality. *PLoS ONE, 8*, e76674

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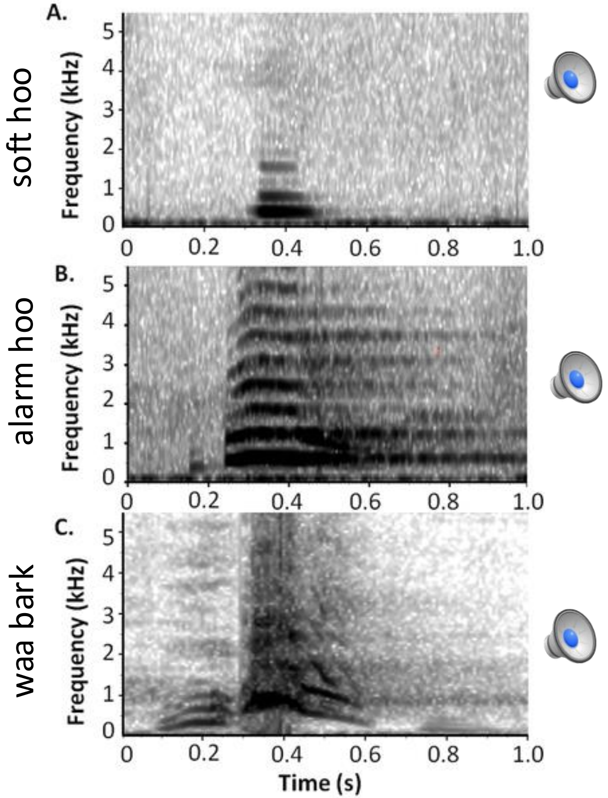

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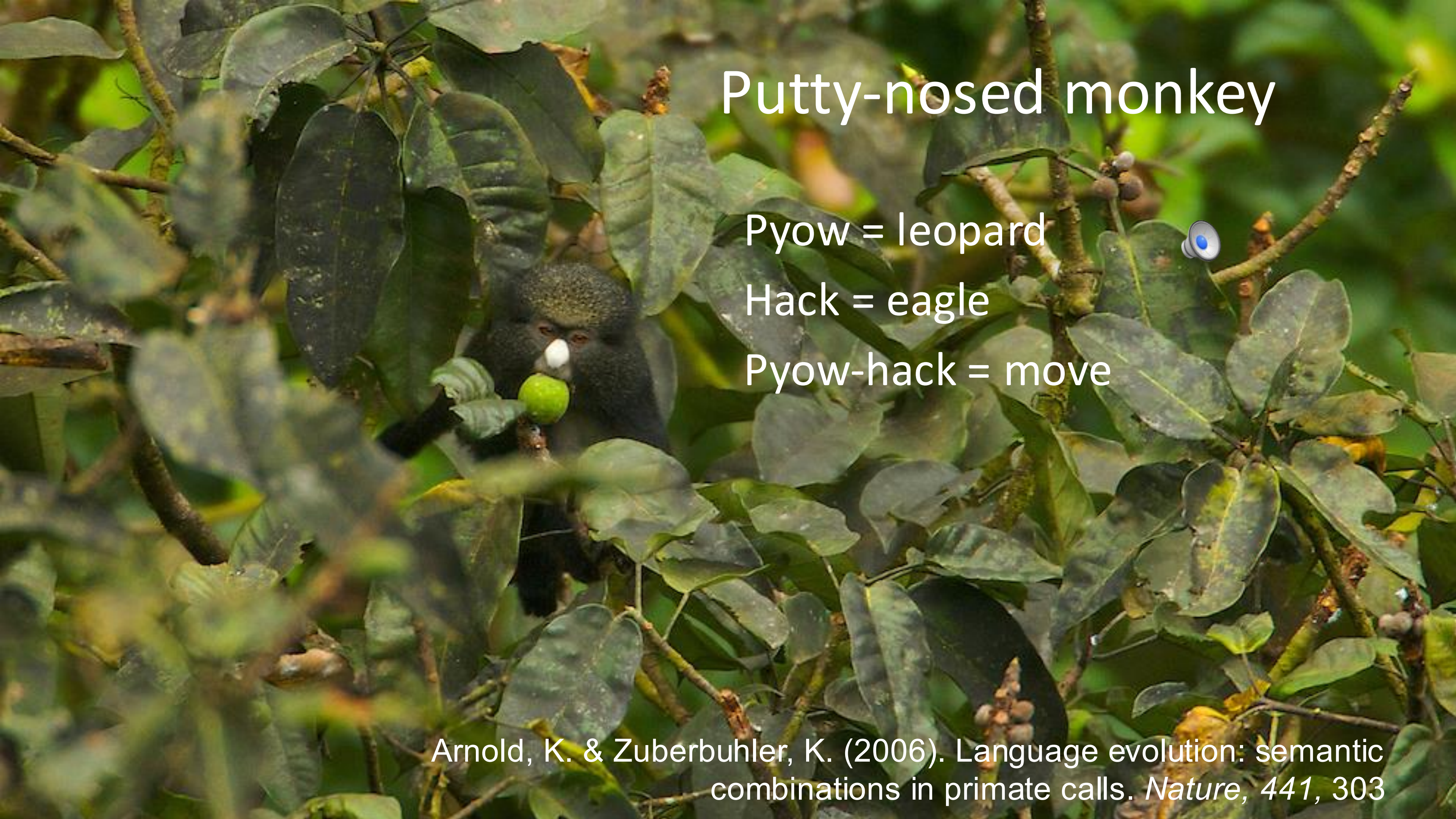
Vocal learning
Learning sequencing constraints
Learning compositional meaning-form mappings



Putty-nosed monkey

Pyow = leopard

Hack = eagle

Pyow-hack = move

Arnold, K. & Zuberbuhler, K. (2006). Language evolution: semantic combinations in primate calls. *Nature, 441,* 303

Campbell’s monkey

Leopard alarm

Eagle alarm

Boom = not urgent

Zuberbühler K (2002) A syntactic rule in forest monkey communication. *Animal Behaviour, 63*, 293–299.

Chimpanzees

Alarm huus: alarm or threat

Waa bark: recruitment call (?)

Alarm huu + waa bark = recruitment signal in a dangerous situation (?)

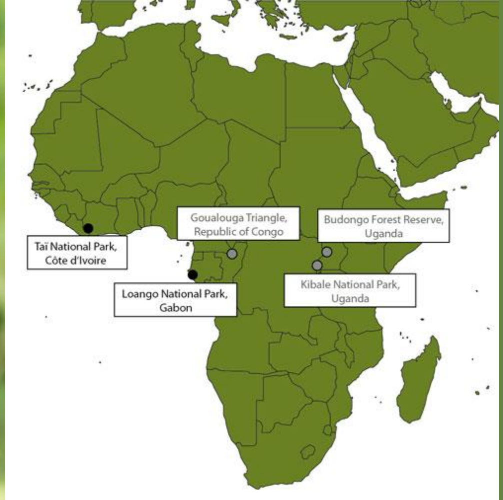
Leroux, M., Schel, A.M., Wilke, C. et al. (2023) Call combinations and compositional processing in wild chimpanzees. *Nature Communications, 14,* 2225.

Extensive and meaningful call combination in chimpanzees and bonobos?

Berthet et al. (2025). Extensive compositionality in the vocal system of bonobos. Science, 388,104-108

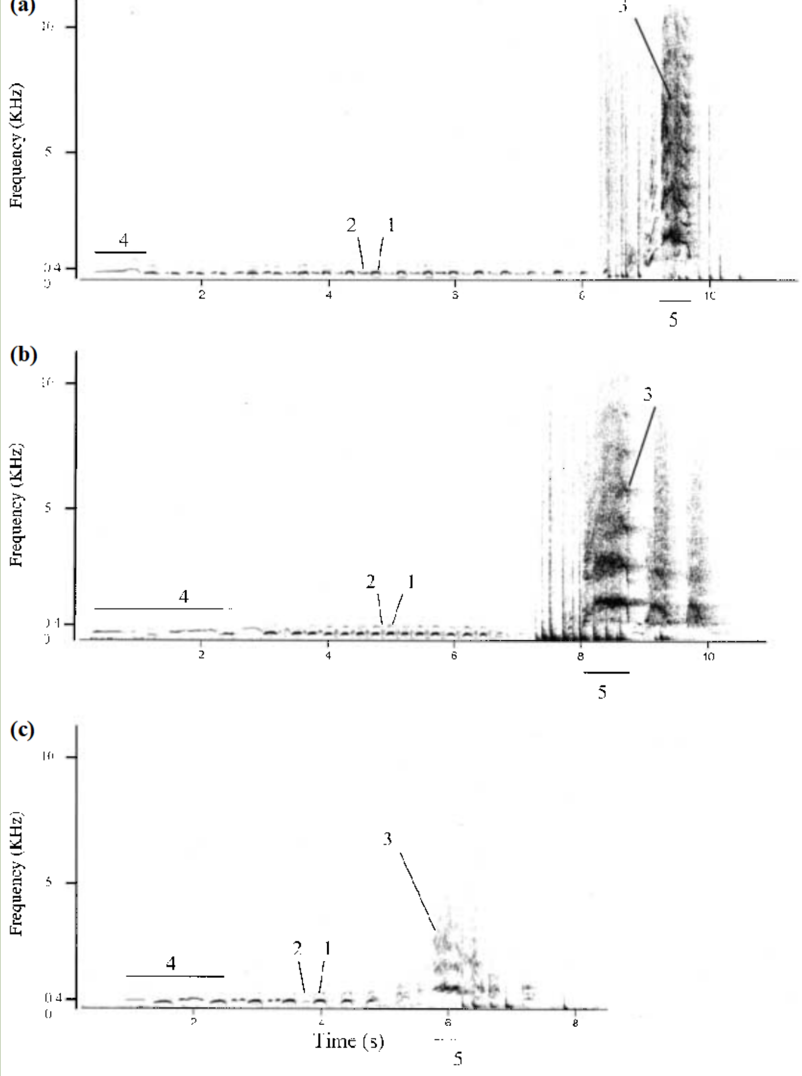
Girard-Buttoz et al. (2025). Versatile use of chimpanzee call combinations promotes meaning expansion. *Science Advances, 11,* eadq2879

Suggestive evidence for learned vocalizations in chimpanzees?



Pant hoots of chimpanzees vary between neighbouring groups

Crockford, C., Herbinger, I., Vigilant, L. & Boesch, C. (2004). Wild Chimpanzees Produce Group-Specific Calls: a Case for Vocal Learning? *Ethology,110,* 221—243.



Crockford et al. (2004): pant hoots of neighbouring groups differ in (e.g.):

- Length of intro (4)
- Peak frequency of screams (3)
- Duration of climax (5)

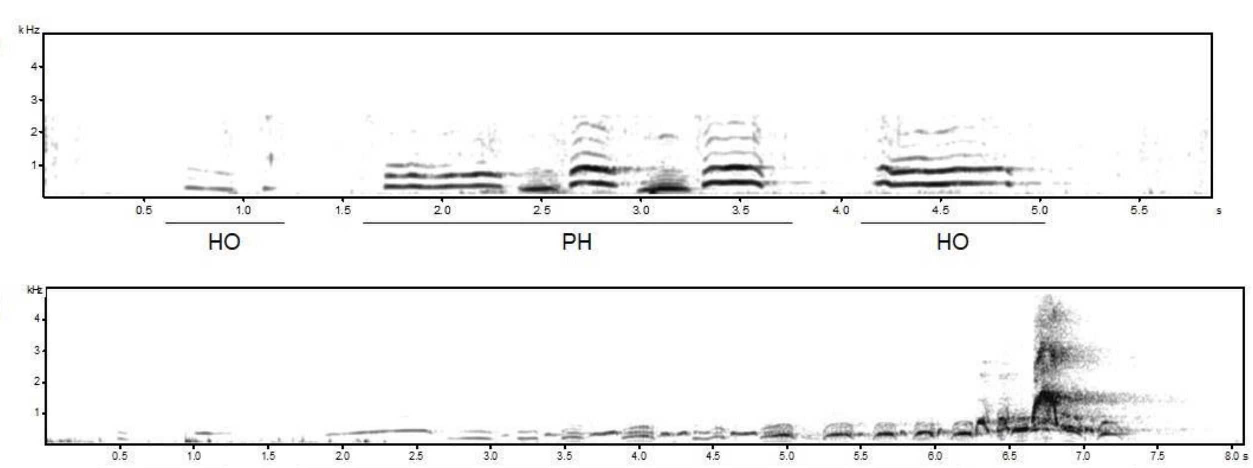
But Desai et al. (2022) fail to replicate in Gombe National Park

- Substantial inter-individual differences, small sample sizes

Crockford, C., Herbinger, I., Vigilant, L. & Boesch, C. (2004). Wild Chimpanzees Produce Group-Specific Calls: a Case for Vocal Learning? *Ethology,110,* 221—243.

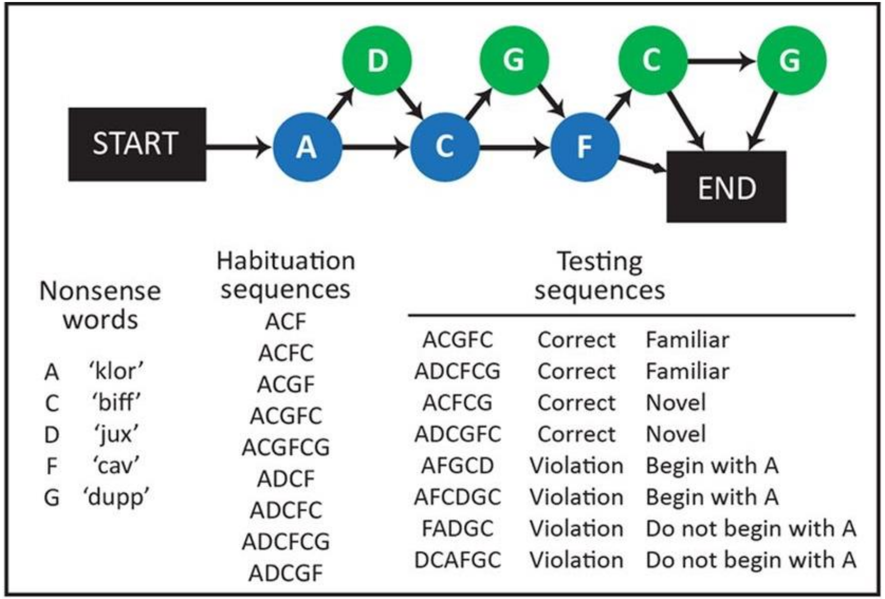
Desai, N. P., Fedurek, P., Slocombe, K. E., & Wilson, M. L. (2022). Chimpanzee pant-hoots encode individual information more reliably than group differences. *American Journal of Primatology, 84,* e23430.

A lot is not known about call combinations in chimpanzees!

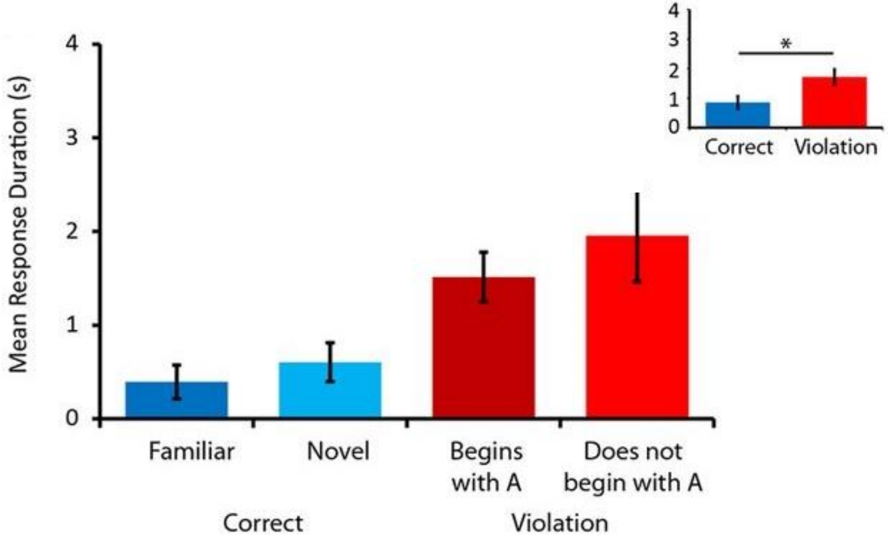


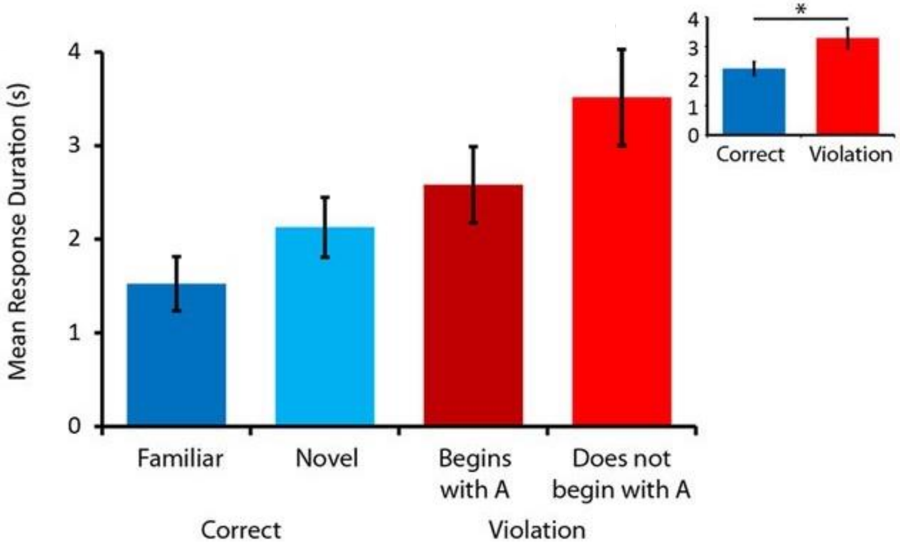
Girard-Buttoz, C., Zaccarella, E., Bortolato, T., Friederici, A. D., Wittig, R. M., & Crockford, C. (2022). Chimpanzees produce diverse vocal sequences with ordered and recombinatorial properties. *Communications Biology*, *5*, 410.

Artificial Grammar Learning in non-humans



Wilson, B., Slater, H., Kikuchi, Y., Milne, A., Marslen-Wilson, W., Smith, K., & Petkov, C. (2013). Auditory artificial grammar learning in macaque and marmoset monkeys. *Journal of Neuroscience, 33,* 18825-18835.
For review see e.g. Petkov, C. I., & Ten Cate, C. (2020). Structured Sequence Learning: Animal Abilities, Cognitive Operations, and Language Evolution. *Topics in Cognitive Science, 12,* 828– 842.





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Non-adjacent dependency learning
(pel X rud, vot X jic)

Training

pel wadim rud… vot kicey jic… pel feenam rud…
vot wadim jic… vot puser jic… pel puser rud…
vot kicey jic… vot skiger jic… pel kicey rud …

Test

pel wadim rud

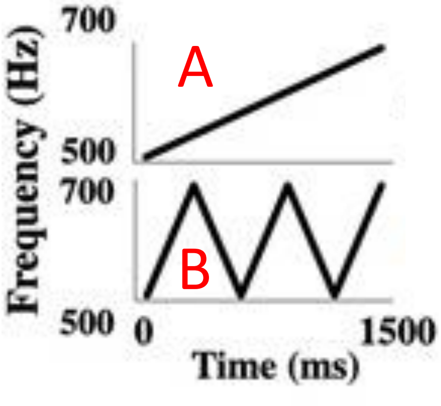
pel wadim jic?

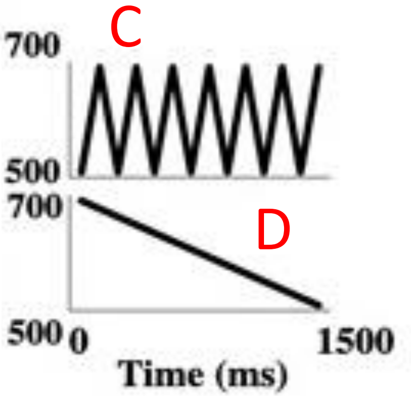
vot wadim jic

vot wadim rud?

Gomez, R. (2002). Variability and detection of invariant structure. *Psychological Science, 13,* 431-436.

Non-adjacent dependency learning (AXB, CXD)





Watson, S. K., Burkart, J. M., Schapiro, S. J., Lambeth, S. P., Mueller, J. L., & Townsend, S. W. (2020). Nonadjacent dependency processing in monkeys, apes, and humans. *Science Advances, 6,* eabb0725.

Non-adjacent dependency learning (AXB, CXD)

Sequence type	Example AD	Example Non-AD
Familiarization	$A_1 – B_7$	$A_1 – X2_4 – B_7$
	$C_5 – D_1$	$C_5 – X1_1 – D_1$
Generalization	$A_{13} – B_{11}$	$A_{13} – X1_{15} – B_{11}$
	$C_{10} – D_{16}$	$C_{10} – X2_{14} – D_{16}$
Violation	$A_{13} – D_{11}$	$A_{13} – X1_{15} – D_{11}$
	$C_{10} – B_{16}$	$C_{10} – X2_{14} – B_{16}$

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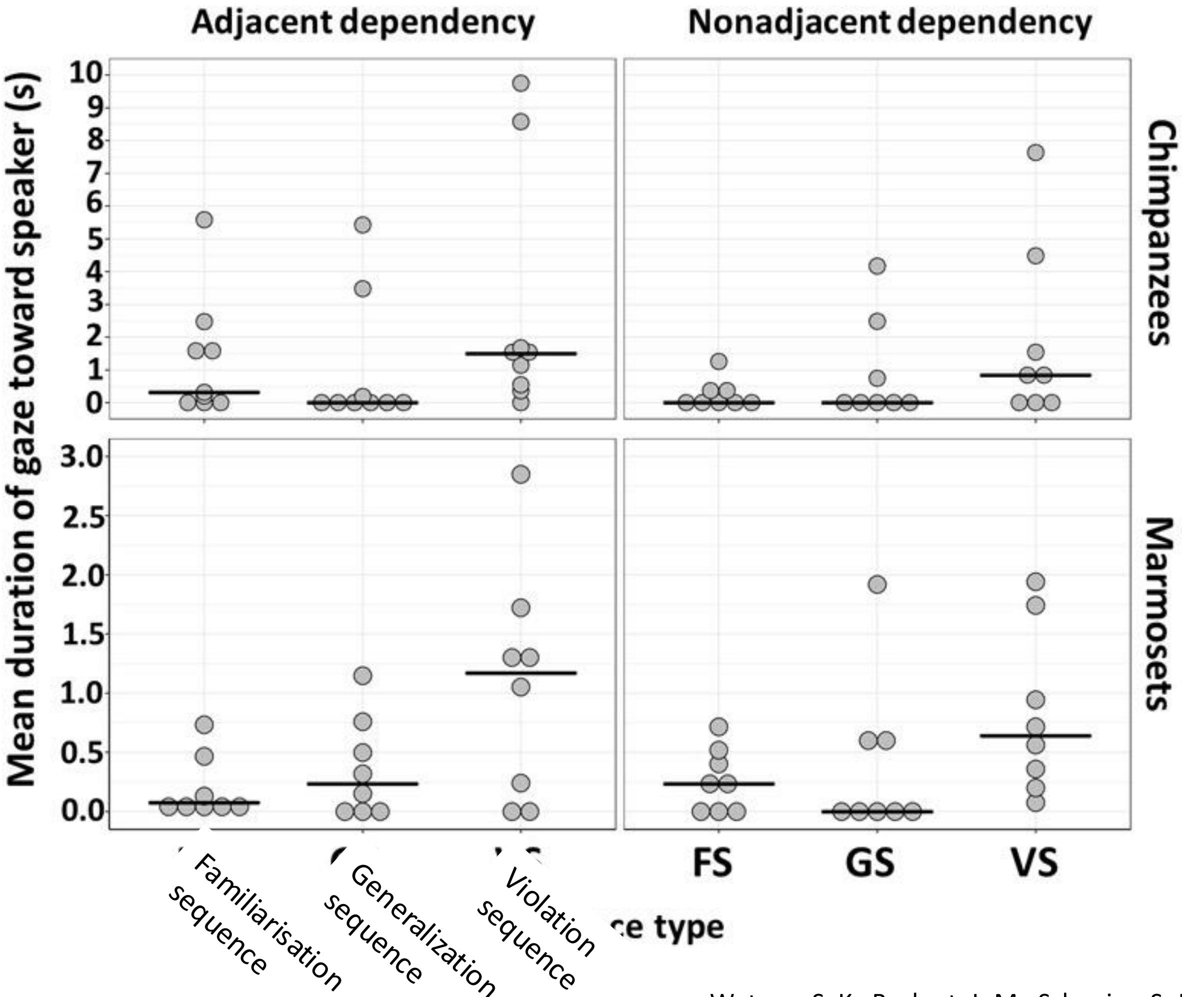

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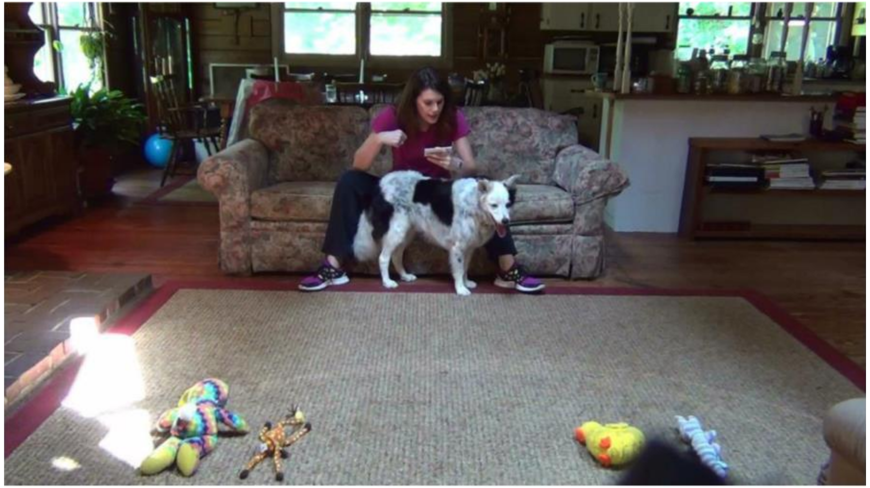
How about learning of **meaningful** sequences?

“ball fetch”
“stick point”

Ramos, D., & Ades, C. (2012). Two-item sentence comprehension by a dog (Canis familiaris). *PLoS ONE, 7,* e29689.



“to sugar take decoy”
“to decoy take sugar”

Pilley, J. W. (2013). Border collie comprehends sentences containing a prepositional object, verb, and direct object. *Learning and Motivation*, *44,* 229-240.

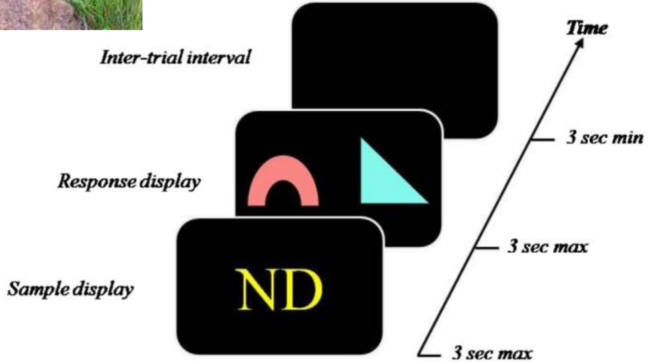
https://youtu.be/2Dhc2zePJFE

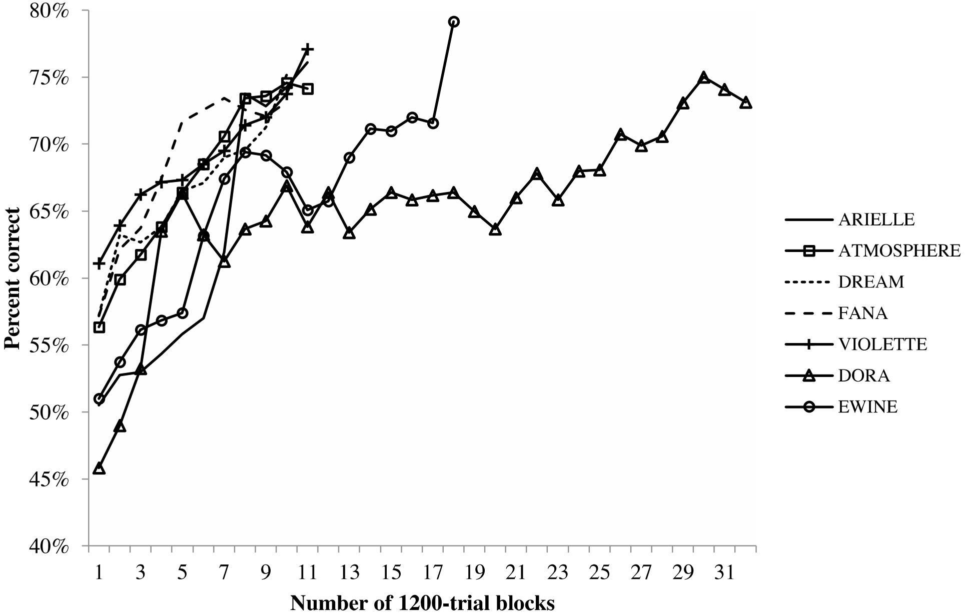
Savage-Rumbaugh, E. S., Murphy, J., Sevcik, R., Brakke, K., Williams, S., Rumbaugh, D., & Bates, E. (1993). Language comprehension in ape and child. *Monographs of the Society for Research in Child Development, 58*, 1–252.

Truswell, R. (2017). Dendrophobia in bonobo comprehension of spoken English. *Mind and Language, 32,* 395-415.

- Could just be ‘semantic soup’ plus smart interpretation?
  - *Cut the onions with your knife*
  - *Put the pine needles in the refrigerator*
- But he can handle reversible events (cf. also Chaser)
  - *Put the tomato in the oil*
  - *Put some oil in the tomato* [Kanzi pours oil in a bowl with the tomato]
- But no strong evidence for **hierarchy**
  - *Give the water and the doggie to Rose*. [Gives dog only]
  - *Give the lighter and the shoe to Rose*. [Gives lighter only]
  - *Give me the milk and the lighter* [Responds correctly]

Puzzling failures in (most) baboons





6 letters (3 for shapes, 3 for colours)
3 shapes, 3 colours

Medam, T., & Fagot, J. (2016). Behavioral assessment of combinatorial semantics in baboons (Papio papio). *Behavior Processes, 123*, 54-62.

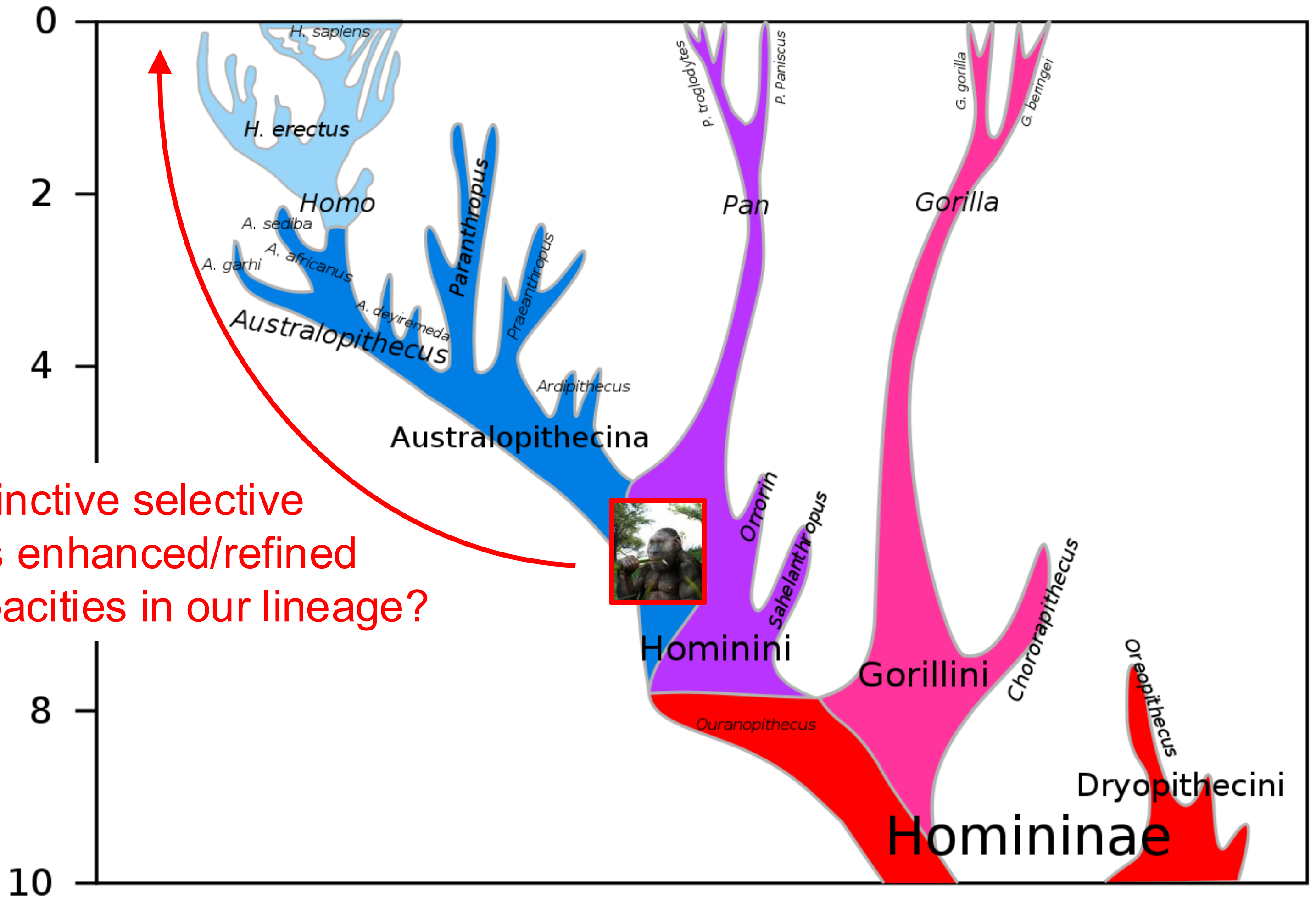
The Last Common Ancestor of chimps and humans

Humans

- The ability to infer and reason about the minds of others
- The ability to learn complex grammars
  - Capacity for complex vocal learning
  - Ability to learn complex sequencing constraints
  - Ability to learn compositional meaning-form mappings

The LCA

- Some ability to infer and reason about the minds of others
- Some ability to learn complex grammars?
  - Little/no capacity for vocal learning?
  - Some ability to learn sequencing constraints?
  - Some ability to learn compositional meaning-form mappings?



What distinctive selective pressures enhanced/refined these capacities in our lineage?

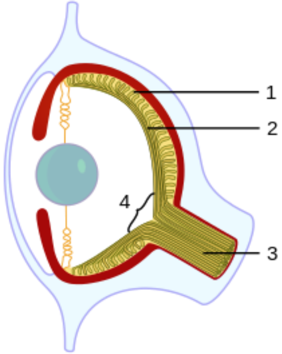
https://commons.wikimedia.org/wiki/File:Hominini_lineage.svg

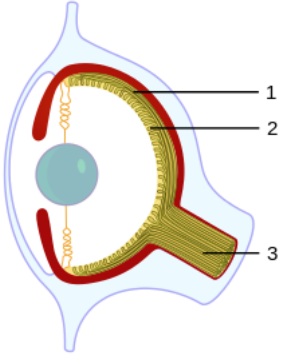
Bonus content – birds!

Convergent evolution










Southern pied babblers

Alert call

Recruitment call

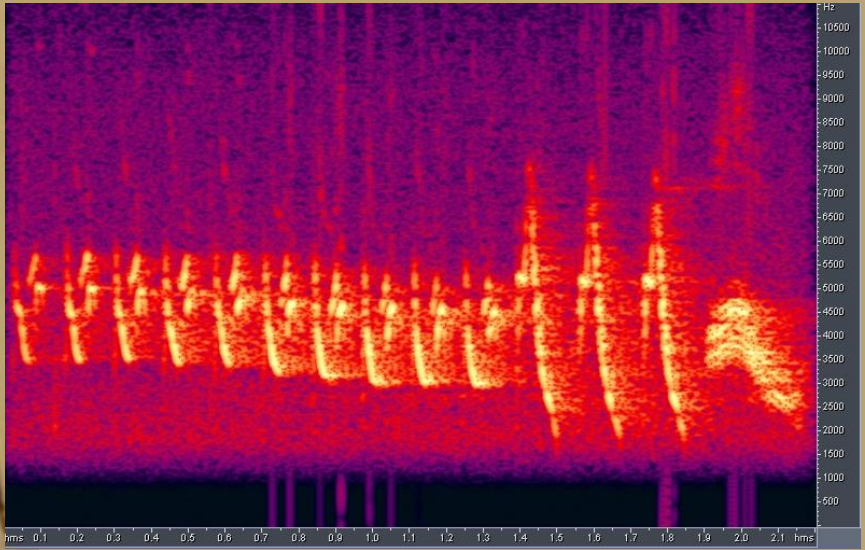
Alert call + recruitment call = mob predator

Engesser, S., Ridley, A. R., & Townsend, S. (2016). Meaningful call combinations and compositional processing in the southern pied babbler. *Proceedings of the National Academy of Sciences, USA, 113*, 5976-5981.

Abundant evidence of structure in **bird song**

- Songs consist of sequences of notes
- Constraints on the order of combination
- Structure in the signal doesn't subserve meaning
- **Vocal learning**
- Ultimate functions
  - Territorial defense
  - Courtship
  - Pair/group bonding (duetting)

Chaffinch song



Structure of chaffinch song (British)

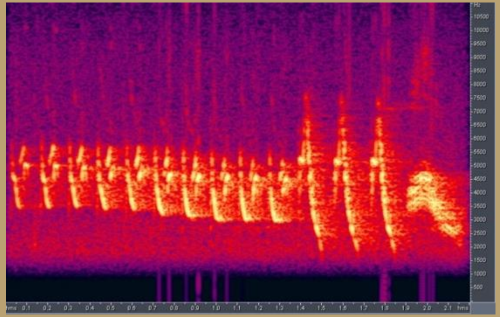
Each bird has 1-6 song types

- Mean 2-3

Order of notes in each song fixed

2-5 trill phrases, followed by a flourish

- Trill: sequence of 2 or more near-identical units
  - Number of repetitions can vary
- Flourish: no repetition
- Transitional notes: single notes between trill phrases
- Re-use of notes
  - Different songs may share, e.g., a flourish



Trill 1

Trill 2

Flourish

Slater, P. J. B., & Sellar, P. J. (1986). Contrasts in the Songs of Two Sympatric Chaffinch Species . *Behaviour, 99,* 46-64.

Slater, P. J. B., Clements, F. A., & Goodfellow, D. J. (1984). Local and regional variations in chaffinch song and the question of dialects. *Behaviour, 88,* 76-97.

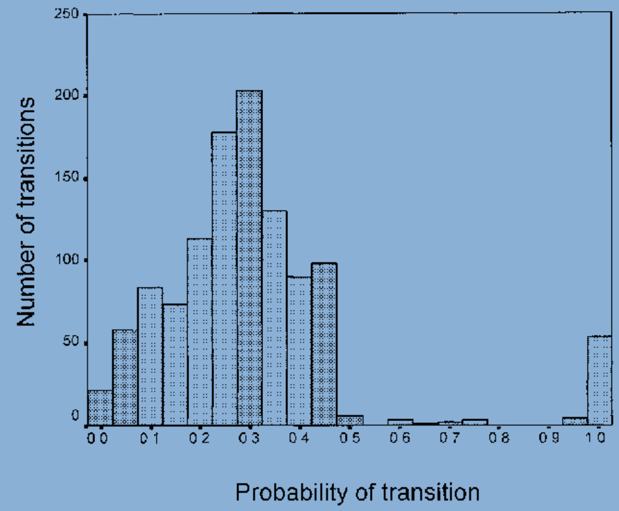
Willow warbler song

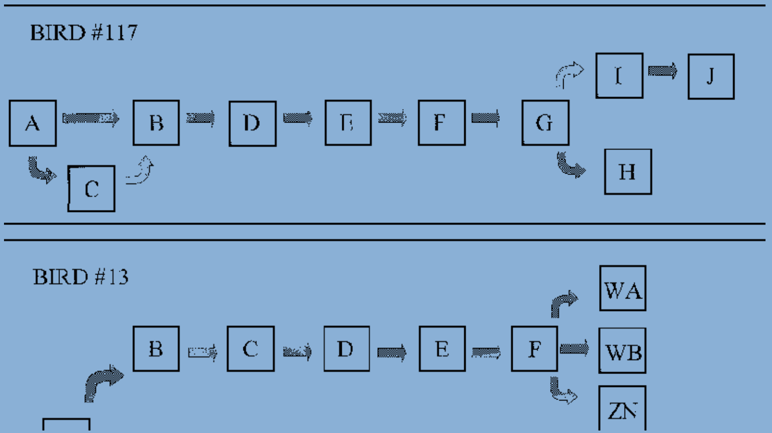
> 100 songs for some birds

- Repertoire size varies

Mix of predictable and less predictable transitions

- A simple grammar

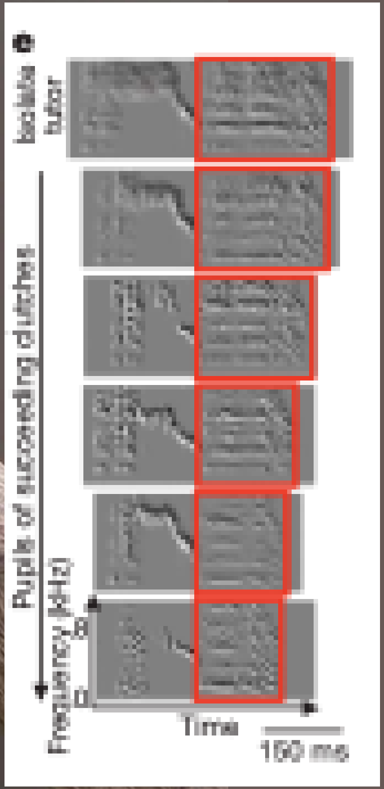




Gil, D., & Slater, P. J. B. (2000). Song organisation and singing patterns of the willow warbler, *Phylloscopus trochilus*. *Behaviour, 137,* 759-782.

Iterated learning in zebra finches

Zebra finch song evolves from isolate to wild-type rapidly (even under quite minimal self-tutoring)



Fehér, O., et al. De novo establishment of wild-type song culture in the zebra finch. *Nature, 459*, 564–568.
Fehér, O., et al. (2017). Statistical learning in songbirds: from self-tutoring to song culture. *Philosophical Transactions of the Royal Society, 372*, B37220160053.